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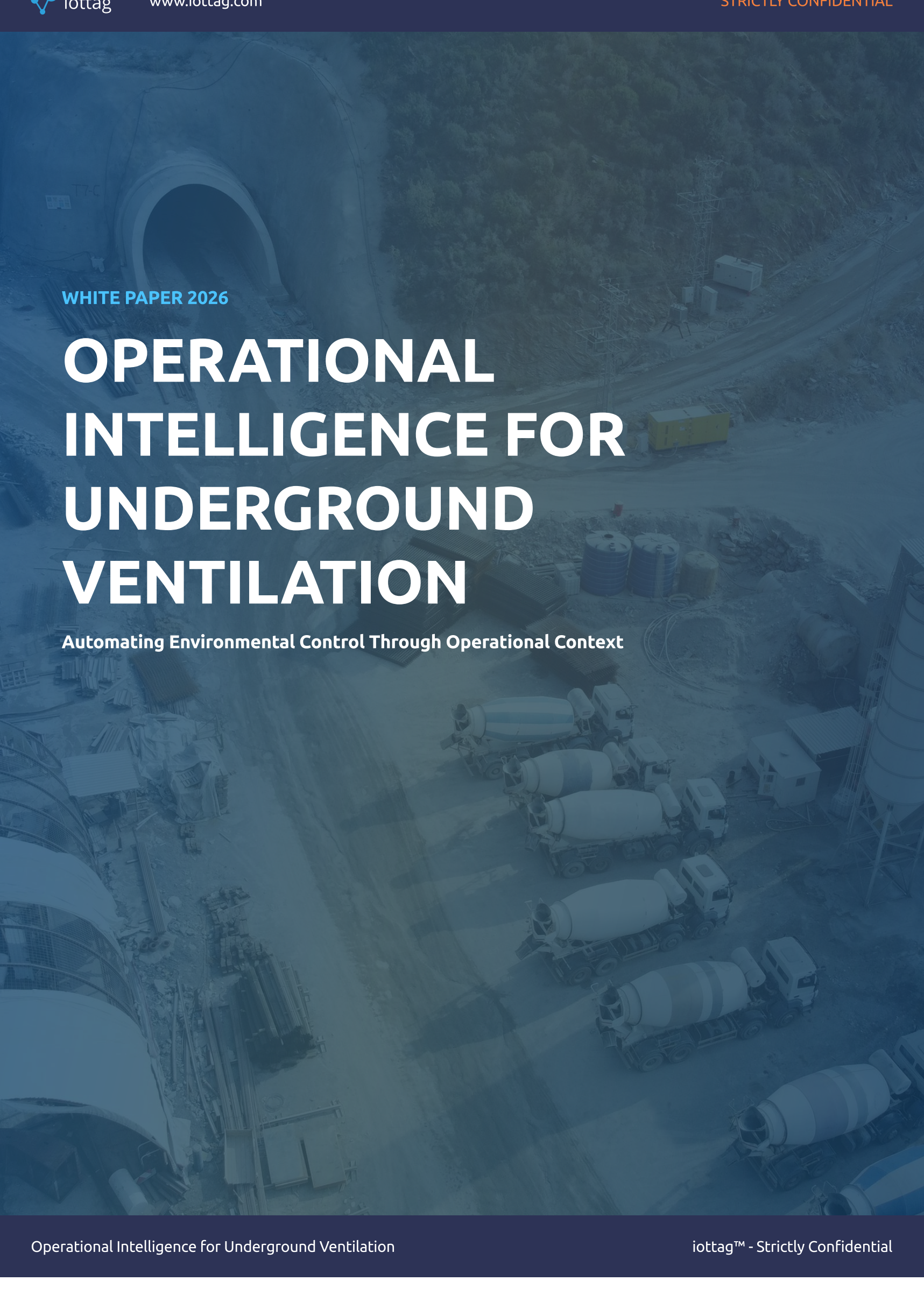


WHITE PAPER 2026

OPERATIONAL INTELLIGENCE FOR UNDERGROUND VENTILATION

Automating Environmental Control Through Operational Context

Executive Summary
Environmental monitoring explains what is happening. Operational Intelligence explains why.

Underground operations have become increasingly connected through environmental monitoring, fleet management, workforce tracking, production systems and industrial control systems. Individually, these technologies provide valuable operational information and contribute to safety, productivity and operational visibility.

Yet each system still operates largely within its own domain:

- Environmental monitoring systems understand air quality.
- Fleet management systems understand vehicle activity.
- Personnel tracking systems understand workforce location.
- Production systems understand output.

Each provides valuable information on its own, but none understands how the operation is performing as a whole. Operational Intelligence introduces a new layer of understanding.

Rather than simply collecting operational data, Operational Intelligence builds a continuously evolving model of how an operation is expected to perform safely, efficiently and productively. Live conditions are compared against this model in real-time to identify operational deviations, understand their broader impact and anticipate emerging risks before they turn into incidents.

Within underground ventilation, this approach turns environmental monitoring from a reactive control system into part of a broader operational decision-making framework. Instead of asking:

> **"What are the environmental conditions?"**

Operations can begin asking:

- Why are conditions changing?
- Which operational activities are contributing?
- What will happen if nothing changes?
- What intervention will provide the greatest operational benefit?

The result is sharper situational awareness, better-informed operational decisions and a stronger balance between safety, productivity and environmental performance.

The Operational Challenge

Modern underground operations generate data. They do not automatically generate understanding.

Modern underground operations rely on hundreds of connected systems, each built to solve a specific operational problem. Examples include:

Operational Domain	Typical Systems	Primary Function
Environment	Ventilation, Gas Monitoring, DPM Monitoring	Environmental safety
Fleet	Fleet Management, Dispatch	Vehicle utilisation
Personnel	Tracking, Access Control	Workforce visibility
Production	Mine Planning, SCADA	Operational performance
Maintenance	Asset Management	Equipment reliability
Safety	Incident, Permit, Compliance	Risk management

Collectively, these systems produce enormous amounts of operational data, but they operate largely independently. Each explains what is happening within its own environment; none explains how the operation is performing collectively. That gap is one of the largest still remaining in operational technology.

An operation may know where every worker is located, current DPM concentration, ventilation status, equipment utilisation and production rates, and still be unable to answer the questions that matter most:

- Is the operation performing as expected?
- Which operational changes are increasing exposure?
- Which systems are influencing one another?
- What risks are beginning to emerge?
- Where should intervention occur first?

These are questions of operational context, not additional data.

The Missing Layer

Most operational systems are designed to monitor assets, people or environmental conditions. Operational Intelligence monitors the operation itself, continuously evaluating how operational domains interact rather than analysing each system in isolation. Examples include:

- How vehicle activity influences DPM.
- How production schedules influence ventilation demand.
- How workforce density influences oxygen consumption.
- How equipment utilisation influences heat generation.
- How environmental conditions influence worker exposure.
- How operational changes influence overall project performance.

Understanding these relationships lets operations move beyond monitoring individual events, toward understanding operational behaviour.

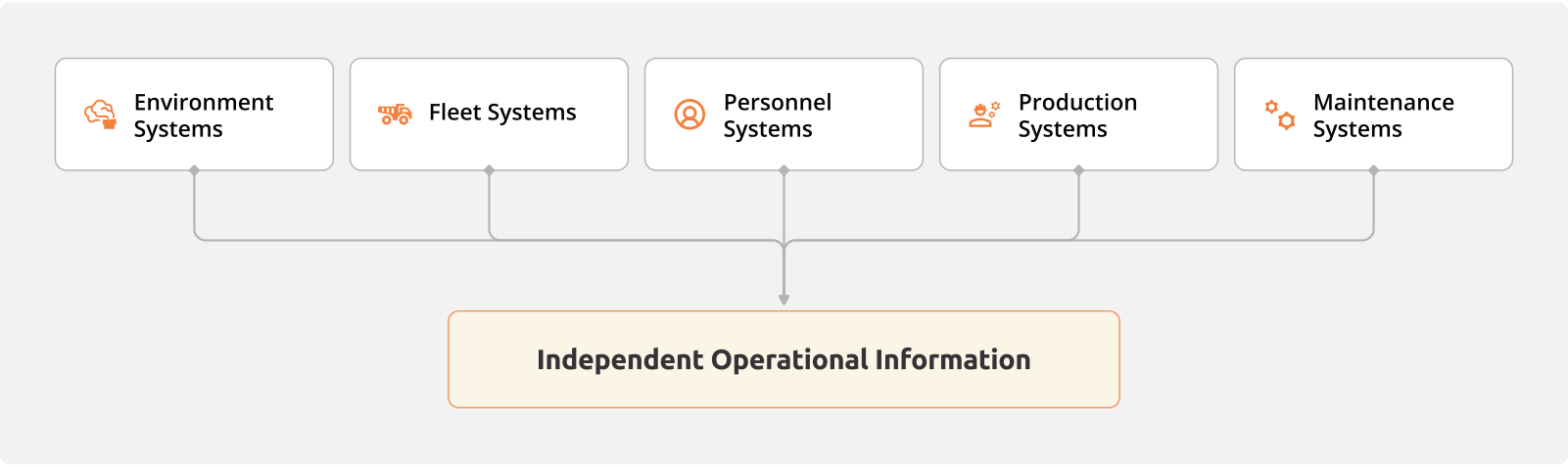


Figure 1 — Traditional Operational Monitoring: independent systems, independent decisions, a disconnected operation.

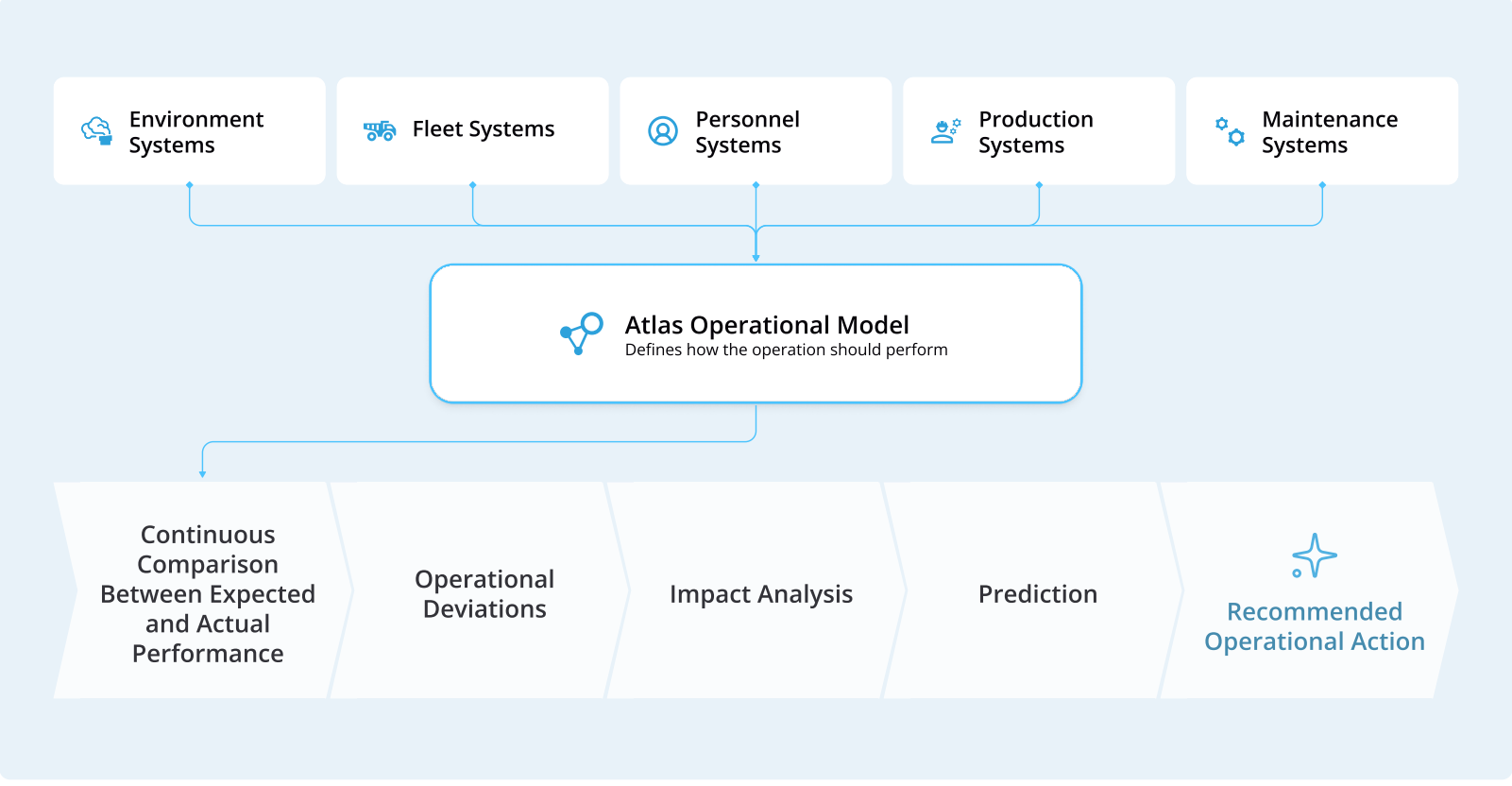


Figure 2 — Operational Intelligence: a single model connects every domain to a continuous cycle of comparison, impact analysis, prediction and recommended action.

From Monitoring to Understanding

Traditional monitoring platforms identify operational events. Operational Intelligence identifies operational behaviour, and that distinction matters. An environmental monitoring platform is able to report that DPM concentrations are increasing; Operational Intelligence explains:

Consider a large infrastructure project.

- Which vehicles contributed to the increase.
- Whether excessive idling is occurring.
- Whether congestion is developing.
- Whether equipment is operating outside expected performance.
- How these changes influence worker exposure.
- What operational intervention is likely to have the greatest effect.

Instead of responding only after environmental conditions change, operations can understand the behaviours creating those conditions and step in earlier, turning monitoring into decision support.

> **Key Principle**
> Individual systems explain what is happening. Operational Intelligence explains how the operation is performing.

Operational Performance Optimisation

Balancing safety, productivity and energy through Operational Intelligence.

Underground operations continuously balance competing objectives. Production teams aim to maximise output. Ventilation systems maintain safe atmospheric conditions. Maintenance teams strive for maximum equipment availability. Safety teams focus on reducing operational risk and worker exposure.

Each objective matters on its own, yet decisions made in one area often ripple through every other part of the operation. Operational Intelligence provides a common framework for understanding and managing these interactions holistically, instead of optimising individual systems in isolation, Atlas helps optimise the performance of the operation as a whole.

Understanding Operational Trade-offs

Operational decisions rarely produce a single outcome. Increasing production can improve output while also raising diesel emissions, ventilation demand and equipment utilisation. Cutting ventilation energy use can lower operating costs but reduce flexibility during periods of peak activity.

Operational Intelligence gives visibility into these interconnected effects before decisions are made. Instead of asking:

> **"Can we increase production?"**

Operations can ask:

- What is the environmental impact?
- Is there enough ventilation capacity available?
- Will worker exposure increase?
- Which operational areas become constrained?
- Can production increase without increasing energy consumption?
- What operational adjustments deliver the best overall outcome?

This shifts decision-making from reactive management to informed planning.

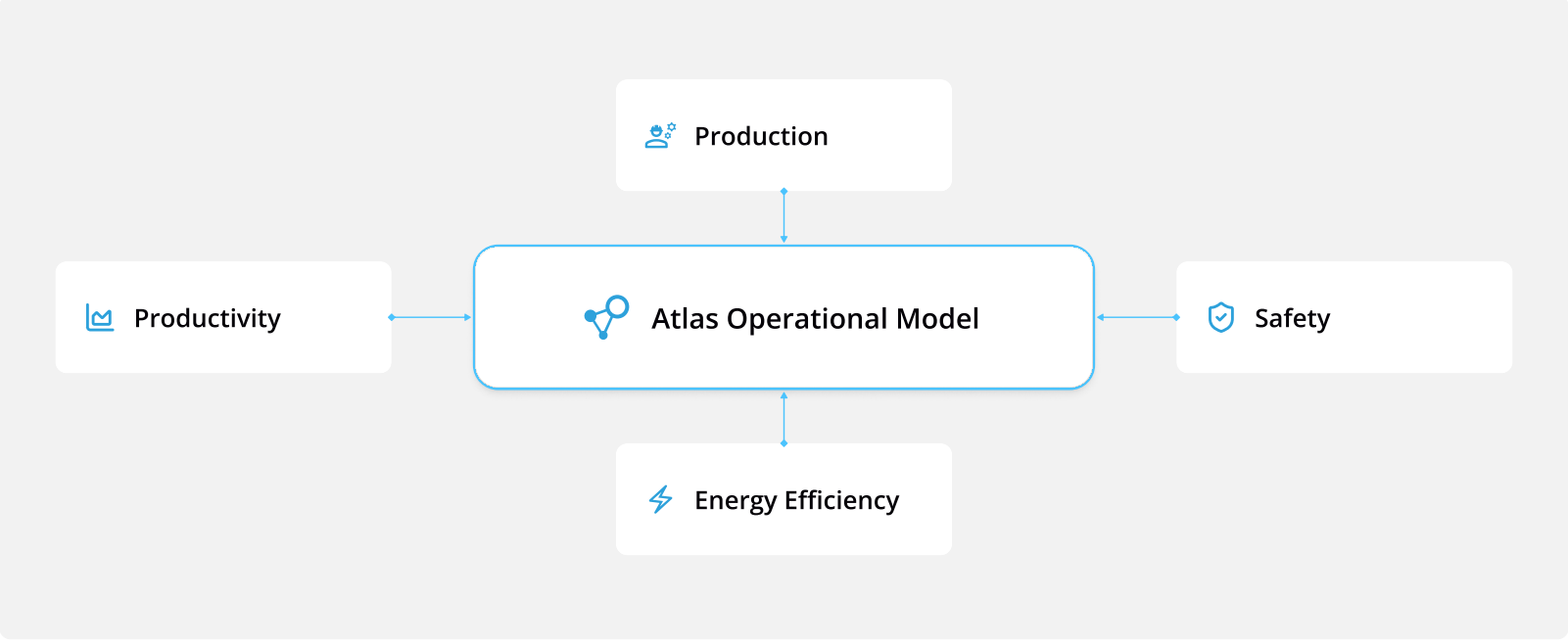


Figure 3 — Balancing Operational Objectives: the Atlas Operational Model sits at the centre of production, safety, productivity and energy efficiency.

The Atlas Operational Model acts as the decision-making layer that evaluates how a change in one operational objective affects the others.

Optimising Ventilation Demand

ariable ventilation systems are highly effective at adjusting airflow to changing environmental conditions. Operational Intelligence complements these systems by helping reduce unnecessary ventilation demand before additional airflow is needed. Examples include:

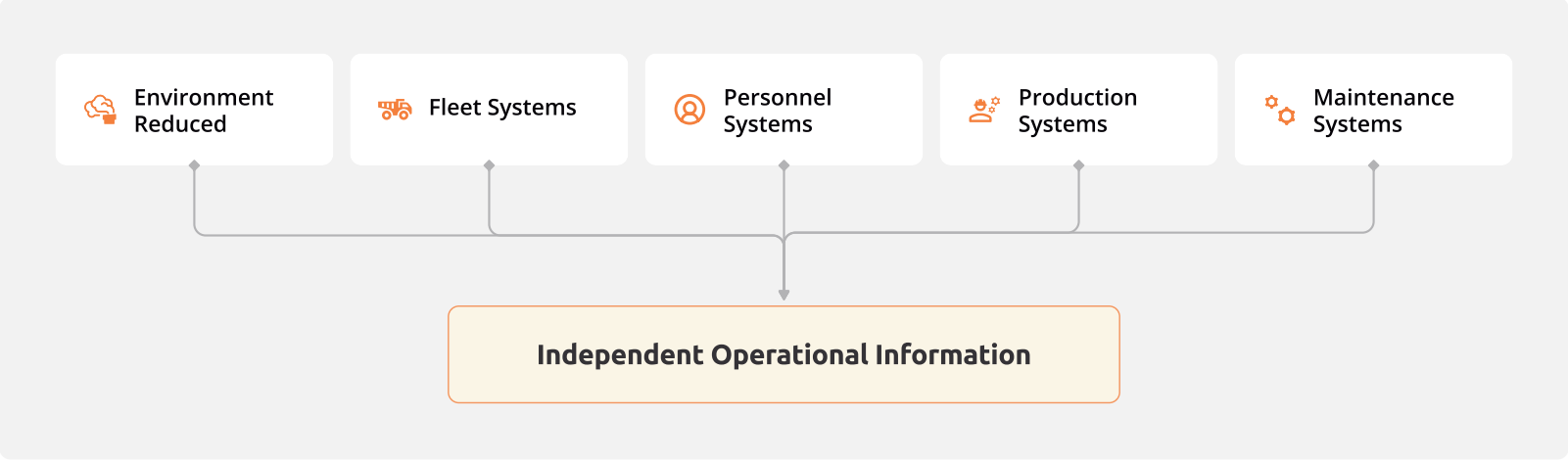
- Eliminating avoidable vehicle idling.
- Improving traffic flow.
- Redistributing equipment activity.
- Optimising work sequencing.
- Scheduling maintenance during lower-demand periods.
- Reducing congestion around active work fronts.
- These improvements reduce the environmental load ventilation systems must manage, improving energy efficiency without compromising worker safety.

Fleet Productivity and Environmental Efficiency

Fleet productivity and environmental performance are closely linked. Every unnecessary vehicle movement burns fuel, generates heat, adds emissions and places extra demand on underground ventilation. Operational Intelligence helps identify inefficiencies such as:

- Repeated haul-route congestion.
- Equipment waiting to load or unload.
- Excessive idle time.
- Underutilised fleet assets.
- Poor dispatch sequencing.
- Unbalanced equipment utilisation.

Addressing these inefficiencies improves productivity while reducing environmental demand at the same time, evidence that operational efficiency and environmental performance are often aligned rather than in competition.



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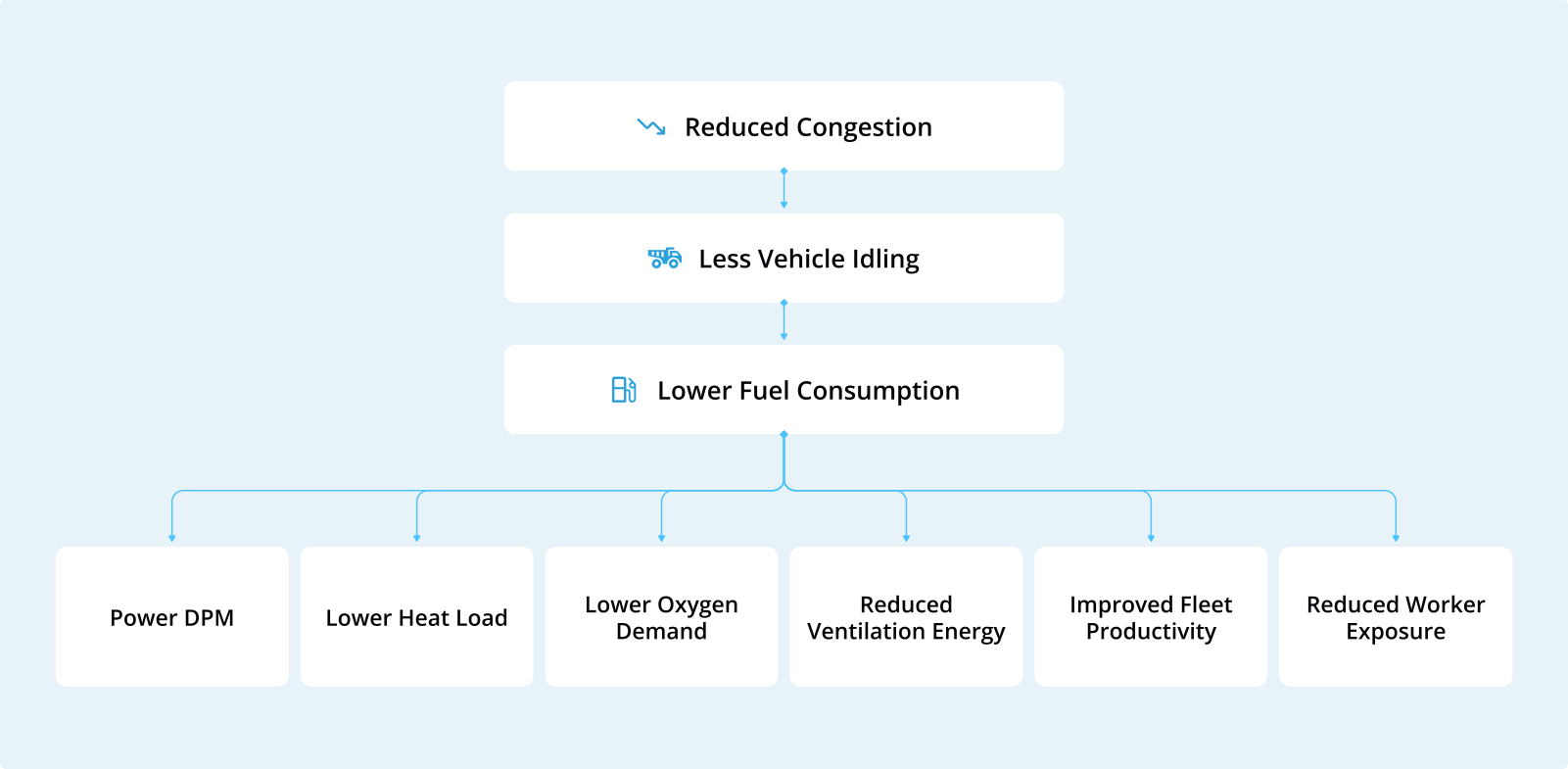


Figure 4 — One Operational Improvement, Multiple Benefits: reducing congestion cascades into gains across six performance indicators.

A single operational improvement can create benefits across multiple performance indicators — the advantage of understanding operational relationships rather than isolated metrics.

Energy Optimisation Through Operational Intelligence

Ventilation is one of the largest energy consumers in underground mining. While variable ventilation systems optimise airflow delivery, Operational Intelligence can help optimise the operational activities that create ventilation demand in the first place.

Examples include:

- Identifying operational practices that unnecessarily increase environmental load.
- Coordinating high-emission activities to improve ventilation efficiency.
- Understanding which work fronts consistently need extra airflow.
- Reducing unnecessary diesel operation within constrained ventilation districts.
- Improving utilisation of existing ventilation infrastructure.

The goal is not simply to cut fan energy, but to make more efficient use of ventilation capacity across the operation.

Predictive Operational Planning

The greatest long-term opportunity lies in moving from reactive operational management to predictive planning. Because Atlas continuously tracks relationships between people, equipment, production and environmental conditions, it can flag emerging trends before they become operational constraints. Potential applications include:

- Forecasting ventilation demand for planned production activities.
- Identifying future congestion based on shift plans.
- Predicting cumulative worker exposure throughout a shift.
- Estimating the environmental impact of introducing additional diesel equipment.
- Evaluating operational scenarios before implementation.

This lets operational planning become proactive instead of reactive.

Operational Objective	Traditional Approach	Atlas Operational Intelligence
Increase production	Add more equipment	Evaluate available environmental and operational capacity before deployment
Reduce energy consumption	Reduce fan speed or operating hours	Reduce unnecessary ventilation demand through improved operational efficiency
Improve fleet performance	Measure utilisation	Analyse movement patterns, idle behaviour and operational constraints
Reduce worker exposure	Respond to environmental alarms	Predict cumulative exposure and optimise work allocation
Improve ventilation performance	Increase airflow	Reduce emissions at their operational source

This shows that Operational Intelligence complements existing technologies by improving the decisions that determine how effective they are.

Engineering Perspective

The future of underground operations is unlikely to hinge on a single technology. Instead, competitive advantage will come from understanding how operational systems interact and making better decisions across them. Operational Intelligence provides the framework for making that possible, rather than treating safety, productivity, energy efficiency and environmental performance as competing priorities, Atlas lets them be managed as interconnected operational objectives.

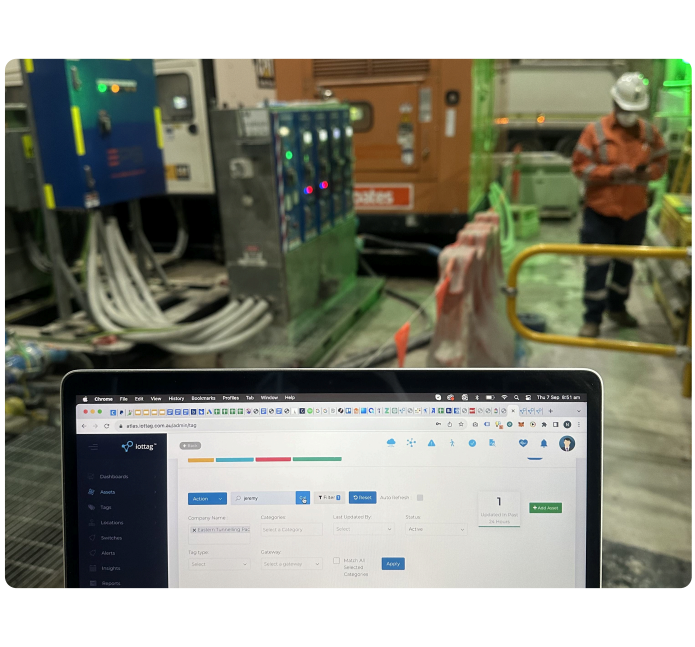
> **Key Principle**
> The greatest gains in underground performance often come not from adding more infrastructure, but from improving how existing people, equipment and systems work together. Operational Intelligence provides the insight needed to make those improvements with confidence.

Partnership Architecture

Connecting environmental intelligence with Operational Intelligence.

Underground mining operations depend on a diverse ecosystem of specialist technologies, each developed to perform a specific operational function, whether monitoring environmental conditions, managing fleet movements, tracking personnel or controlling infrastructure. These systems are highly capable within their respective domains and form the foundation of modern underground operations.

Operational Intelligence is not intended to replace these technologies. Instead, it adds the contextual layer that lets information from multiple operational systems be understood collectively, supporting better decisions across the mine.



Every Platform Has a Different Role

A modern underground operation may include technologies responsible for:

- Ventilation modelling and control
- Environmental monitoring
- Fleet management and dispatch
- Personnel tracking
- Production planning
- Maintenance management
- SCADA and process control
- Communications and telemetry
- Asset health monitoring

Each answers an important operational question.

Platform Type	Primary Question Answered
Ventilation Systems	What are the current underground environmental conditions?
Fleet Management	Where are vehicles operating and how are they performing?
Personnel Tracking	Where are workers located?
Production Systems	What work is being completed?
Maintenance Systems	What is the condition of equipment?
SCADA & Control Systems	What is the current operating state of infrastructure?

Operational Intelligence introduces a new question:

> **"How is the operation performing as a whole?"**

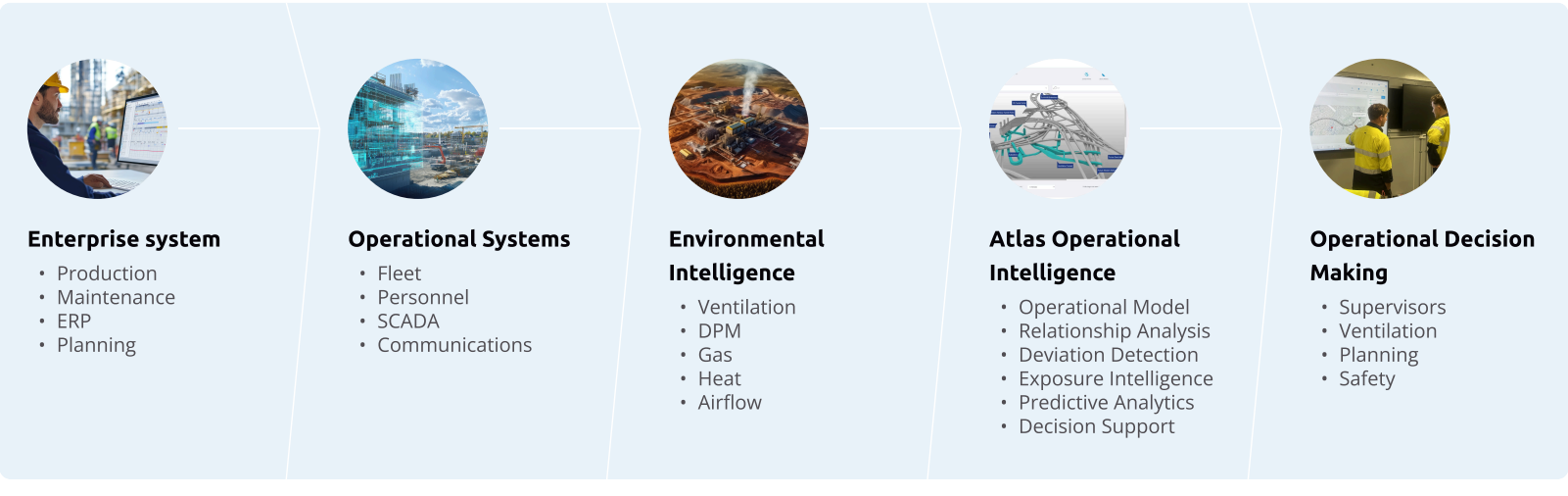


Figure 5 — The Operational Technology Stack: enterprise, operational and environmental systems feed a single Atlas Operational Intelligence layer that supports decision-making.

Creating More Value From Existing Investments

Mining organisations have already invested heavily in operational technology. The challenge is rarely a lack of data, it is understanding how that data relates across systems. Operational Intelligence increases the value of existing technology investments by providing:

- Cross-system operational visibility.
- Contextual understanding of operational events.
- Early identification of emerging operational issues.
- Predictive operational insight.
- Decision support across multiple operational disciplines.

Rather than replacing specialist platforms, Atlas strengthens their collective effectiveness.

A Partnership Opportunity

For ventilation providers, Operational Intelligence is a chance to extend the value of environmental systems beyond measurement and control. Environmental intelligence becomes one of the key inputs into a broader operational model, and together the two technologies give a richer picture of underground operations.

For example, environmental systems identify:

- Increasing DPM
- Reduced oxygen
- Elevated heat
- Changing airflow

Operational Intelligence adds further context:

- Which operational activities caused the change.
- Which equipment contributed most.
- Which workers may be affected.
- Whether the change is temporary or developing.
- Which operational intervention is likely to be most effective.

The result is a more informed, proactive operational response.

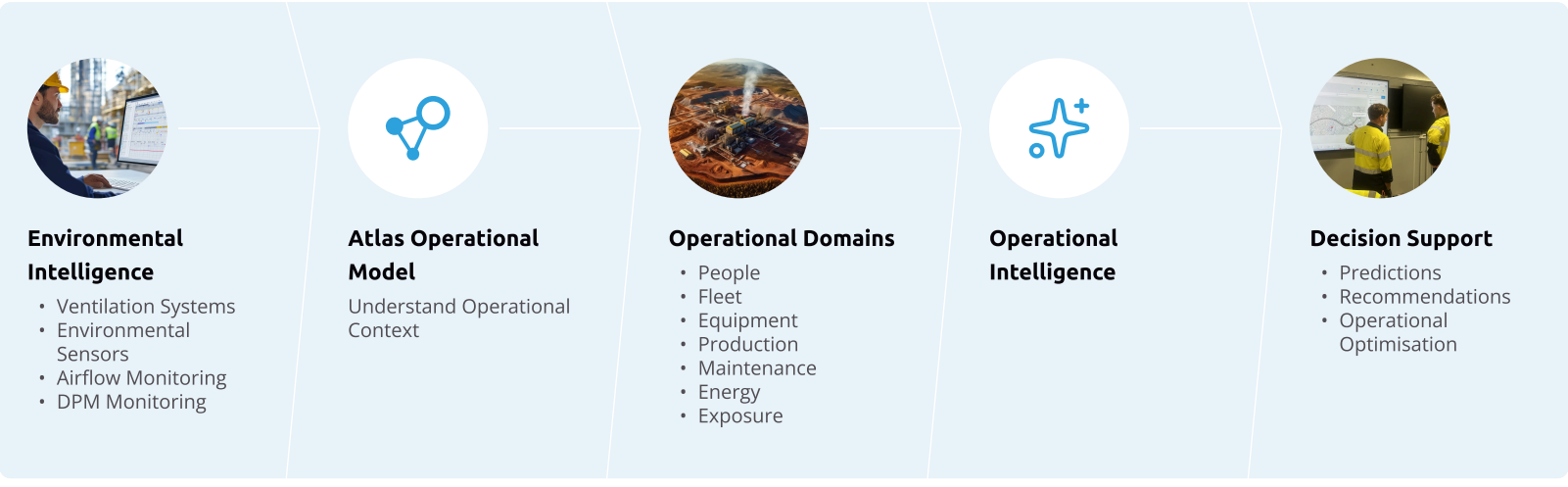


Figure 6 — Partnership Architecture: environmental intelligence feeds the Atlas Operational Model, which reasons across every operational domain to produce decision support.

Shared Outcomes

Combining Environmental Intelligence with Operational Intelligence lets mining organisations achieve outcomes that neither platform could deliver on its own.

Environmental Intelligence	Operational Intelligence	Combined Outcome
Measures environmental conditions	Explains operational causes	Faster identification of root causes
Controls airflow	Reduces unnecessary ventilation demand	Lower energy consumption
Monitors atmospheric safety	Predicts cumulative exposure	Improved workforce protection
Detects environmental change	Predicts operational consequences	Earlier intervention
Maintains compliance	Optimises operational behaviour	Improved safety and productivity

This complementary approach strengthens both technologies without changing what either is fundamentally for.

Operational Intelligence in Practice

Applying the Atlas Operational Model to real operational challenges.

The principles of Operational Intelligence become most valuable when applied to everyday operational decisions. The scenarios below show how environmental information and operational context can combine to support safer, more productive and more energy-efficient underground operations.

These examples are illustrative, meant to demonstrate the methodology rather than prescribe operational procedures.

Traditional Response
Environmental monitoring identifies increasing DPM concentrations within a haulage drive.
Variable ventilation increases airflow to maintain safe atmospheric conditions.
Operations continue: The environmental condition is managed, but the operational cause remains unchanged.

Operational Intelligence Response
Atlas identifies:

- Three haul trucks waiting at an intersection.
- Excessive idle time over the previous twenty minutes.
- A loader operating below expected productivity.
- Increasing queue length.
- Workers entering the affected ventilation district.

Instead of simply increasing airflow, supervisors can:

- Redistribute haul traffic.
- Improve the loading sequence.
- Reduce idle time.
- Restore normal traffic flow.

The environmental condition improves because the operational cause has been addressed.

✓ Lower DPM generation ✓ Lower ventilation demand ✓ Improved fleet productivity
✓ Reduced fuel consumption ✓ Reduced worker exposure

This complementary approach strengthens both technologies without changing what either is fundamentally for.

Scenario 2 — Identifying an Underperforming Vehicle

Combining Environmental Intelligence with Operational Intelligence lets mining organisations achieve outcomes that neither platform could deliver on its own.

Traditional Response
Environmental monitoring detects gradually increasing DPM concentrations across multiple shifts, with no obvious environmental fault identified.

Operational Intelligence Response
Atlas compares emission behaviour across similar equipment and finds that one haul truck consistently produces higher emissions under comparable operating conditions. Maintenance history points to recent injector servicing, so the equipment is flagged for inspection before environmental performance deteriorates further.

✓ Earlier maintenance intervention ✓ Reduced emissions ✓ Improved fleet reliability
✓ Lower ventilation demand ✓ Reduced operating costs

This complementary approach strengthens both technologies without changing what either is fundamentally for.

Scenario 3 — Managing Ventilation Capacity

Traditional Response
A new production heading is scheduled, and additional ventilation capacity is allocated as a precaution.

Operational Intelligence Response
Atlas evaluates:

- Planned diesel equipment.
- Historical activity.
- Expected workforce density.
- Existing ventilation utilisation.
- Environmental capacity.

The analysis shows current ventilation capacity is sufficient, provided equipment arrival is staggered by thirty minutes. No additional infrastructure is required.

✓ Improved utilisation of existing ventilation ✓ Lower energy consumption
✓ Improved production planning ✓ Reduced infrastructure cost

This complementary approach strengthens both technologies without changing what either is fundamentally for.

Scenario 4 — Understanding Worker Exposure

Traditional Response
Environmental monitoring confirms acceptable atmospheric conditions throughout a shift. Operational Intelligence identifies:

- Maintenance personnel remained in a high diesel-activity zone significantly longer than planned.
- Haul traffic increased unexpectedly.
- Worker exposure accumulated despite atmospheric conditions staying within acceptable limits.

Operational Intelligence Response
Atlas recommends:

- Rotating personnel.
- Adjusting maintenance timing.
- Reviewing work sequencing.

✓ Lower cumulative exposure ✓ Improved workforce wellbeing
✓ Better planning ✓ Improved operational awareness

This complementary approach strengthens both technologies without changing what either is fundamentally for.

Scenario 5 — Predicting Tomorrow's Operation

Traditional Response
Analysing yesterday's performance

Operational Intelligence Response
Atlas evaluates tomorrow's planned activities using:

- Production schedule
- Planned fleet deployment
- Maintenance activities
- Ventilation configuration
- Workforce allocation

From this, the Operational Model estimates:

- Expected ventilation demand.
- Potential congestion.
- High-risk operational interactions.
- Predicted exposure hotspots.
- Operational bottlenecks.

Supervisors can adjust the plan before work begins.

This complementary approach strengthens both technologies without changing what either is fundamentally for.

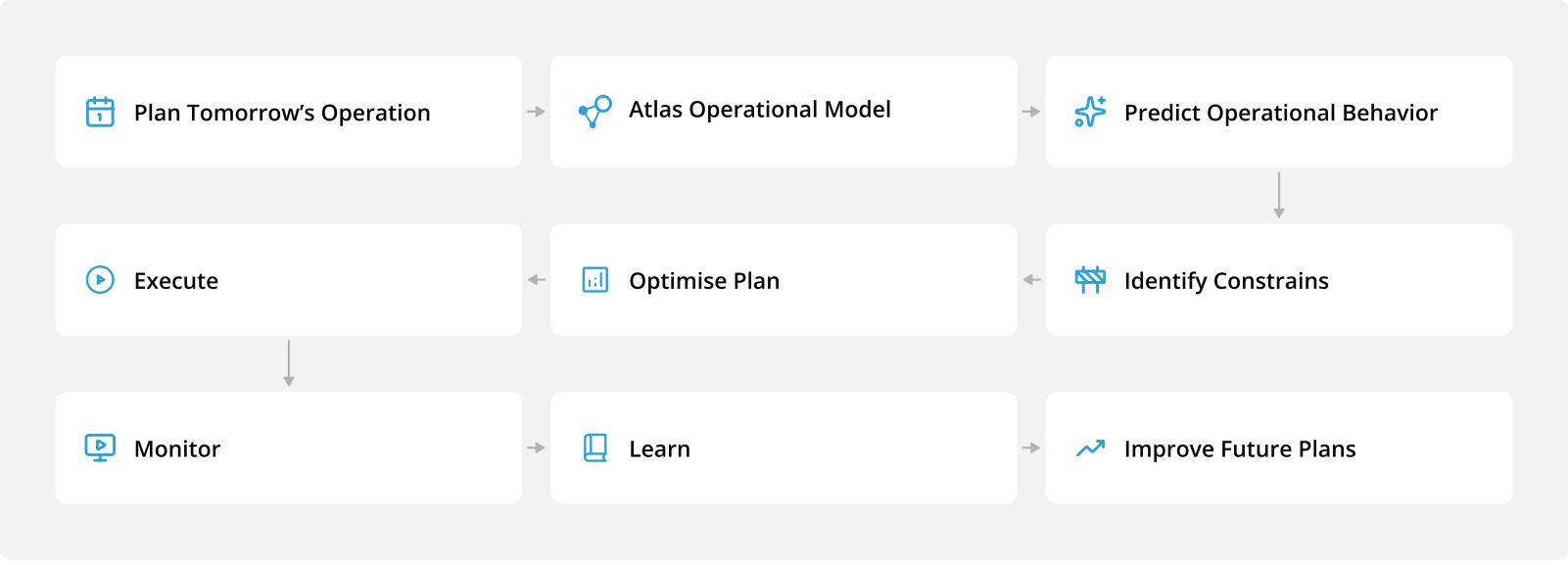


Figure 7 — Operational Intelligence Across the Planning Cycle: plan, predict, execute and learn feed back into the next planning cycle.

Operational Intelligence Is a Continuous Improvement Framework

One defining feature of the Atlas methodology is that it supports every stage of the operational lifecycle.

Phase	Traditional Systems	Atlas Operational Intelligence
Plan	Defines work schedules	Predicts operational performance before execution
Execute	Monitors operational activity	Evaluates performance against the Operational Model
Respond	Reacts to alarms and incidents	Identifies deviations and recommends interventions
Review	Analyses historical data	Learns from operational outcomes to refine future planning

Rather than operating solely as a monitoring platform, Atlas provides a continuous operational improvement framework.

One defining feature of the Atlas methodology is that it supports every stage of the operational lifecycle.

> **Final Key Message**
> Operational Intelligence is not another monitoring system. It is a methodology for understanding, predicting and continuously improving operational performance by connecting people, equipment, environment and production within a single operational model.

Conclusion

From monitoring the mine to understanding the mine.

Underground mining has entered an era of information abundance. Environmental monitoring systems measure atmospheric conditions with increasing precision. Fleet management systems track equipment in real time. Personnel tracking systems provide workforce visibility. Production and maintenance platforms generate detailed operational information.

Yet despite these advances, one question often remains hard to answer:

> **"Is the operation performing as it should?"**

That's the challenge Operational Intelligence addresses. Rather than treating each technology as an isolated source of information, Atlas builds a model of how the operation is expected to perform and continuously compares live conditions against it.

By understanding the relationships between people, equipment, production activities and environmental conditions, Operational Intelligence turns operational data into meaningful decision support.

Within underground ventilation, this creates a powerful partnership. Environmental Intelligence continues to provide accurate measurement, modelling and control of underground conditions, while Operational Intelligence adds the context needed to understand why those conditions exist, how they are evolving and what actions are most likely to improve outcomes. Together, these capabilities support a safer, more productive and more energy-efficient operation.

The future of safe and efficient underground mining requires the unification of specialist technologies to unlock true understanding of operational performance. Atlas Operational Intelligence provides that connecting layer.

Explore the future of operational intelligence

Read the latest perspectives, research and industry insights from the iottag team